

HYDROGENATED NBR SYNTHESIS AND PROCESS DEVELOPMENT: HYDROGENATION, METATHESIS, AND LOW-MOONEY ROUTE PATENTS

1. ABSTRACT

This report synthesizes a set of primary patents addressing hydrogenated nitrile rubber (HNBR) preparation, process intensification, and property control. The target material is HNBR derived from nitrile butadiene rubber (NBR), with emphasis on reduced residual unsaturation, improved heat/oil resistance, and elastomeric performance. Across the selected patents, the landscape spans three major technical directions: direct hydrogenation of NBR in solution or specialized reactors, simultaneous metathesis/hydrogenation to lower molecular weight before or during saturation, and post-hydrogenation modification or downstream use in blends and compositions. The patents collectively define HNBR as a partially or fully hydrogenated nitrile rubber with residual double bond content typically in the low single-digit to teens range, while also highlighting the importance of controlling Mooney viscosity, molecular weight, and gel formation.

The dominant polymerization/synthesis approaches include homogeneous hydrogenation with noble-metal catalysts such as rhodium, ruthenium, palladium, and cobalt-based systems; heterogeneous hydrogenation using supported catalysts; and metathesis-assisted routes using ruthenium/osmium carbene catalysts or related catalyst systems. Several patents focus on process engineering rather than catalyst chemistry, including spinning disk reactor hydrogenation and catalyst recovery/removal strategies to improve product aging and color. Others emphasize simultaneous hydrogenation and metathesis to achieve low molecular weight HNBR in a single step, with co-olefin-assisted chain scission and narrow molecular weight distributions. A separate patent addresses reduction of nitrile groups in already hydrogenated rubber using Raney cobalt, showing that HNBR can be further functionalized to amino-containing polymers.

Formulation and process trends include solution-phase operation in chlorobenzene or THF/dioxane mixtures, hydrogen pressures from sub-MPa to tens of MPa, temperatures from about 60 to 220°C depending on route, and catalyst loadings reported in phr or ppm. The evidence consistently shows that catalyst choice, catalyst residue control, and reaction-time management are central to balancing hydrogenation degree, molecular-weight retention, and final HNBR performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2580260A1
- Patent Title: Blends of partially hydrogenated nitrile rubber and silicone rubber, ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of rubber blends based on partially hydrogenated nitrile rubber and vinyl-group-containing silicone rubber. The key process is mixing the two rubbers in the presence of a metathesis catalyst, specifically a complex catalyst based on a metal of the 6th or 8th transition group with at least one carbene-like ligand, followed by vulcanization of the blend. - Organic solvent content: at most 1200 ppm — Claim 2 - Organic solvent content: at most 500 ppm — Claim 2 - Organic solvent content: at most 100 ppm — Claim 2

Relevant Experimental Evidence - Example 1: demonstrates dynamic crosslinking of partially hydrogenated nitrile rubber with silicone rubber using hydrosilylation and subsequent conventional crosslinking. - Example 1: shows that prior-art blends can suffer from inadequate low- and high-temperature mechanical properties. - Example 1: illustrates the use of metathesis catalysts to improve compatibility in HNBR/silicone rubber blends.

Technical Relevance This patent is relevant because it uses partially hydrogenated nitrile rubber as a key blend component and shows how HNBR can be incorporated into high-performance elastomer systems. Although it does not prepare HNBR itself, it is important for downstream processing and application of hydrogenated NBR materials.

Patent 2

- Patent Number: EP3116916B1
- Patent Title: Process for the reduction of nitrile groups of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a post-hydrogenation functionalization process in which partially or completely hydrogenated nitrile rubber is contacted with Raney cobalt in hydrogen to reduce nitrile groups to amino groups. The process is designed to tune the degree of nitrile-group reduction while preserving the hydrogenated rubber backbone. - Weight-average molecular mass Mw: 20,000 to 1,000,000 — Claim 1 - Weight-average molecular mass Mw: 100,000 to 800,000 — Claim 1 - Weight-average molecular mass Mw: 200,000 to 500,000 — Claim 1 - Catalyst loading: 0.01 to 5.0 g catalyst per g rubber — Claim 1 - Catalyst loading: 0.07 to 3 g catalyst per g rubber — Claim 1 - Catalyst loading: 0.1 to 2 g catalyst per g rubber — Claim 1 - Catalyst loading: 0.2 to 1 g catalyst per g rubber — Claim 1 - Reaction time: 1 to 72 h — Claim 1 - Reaction time: 2 to 36 h — Claim 1 - Reaction time: 4 to 24 h — Claim 1 - Hydrogen pressure: 0.5 to 25 MPa — Claim 1 - Hydrogen pressure: 1.5 to 18 MPa — Claim 1 - Hydrogen pressure: 3 to 15 MPa — Claim 1 - Hydrogen pressure: 5 to 12 MPa — Claim 1 - Reaction temperature: 170 to 220 degrees C — Claim 1 - Nitrile-group reduction limit: not more than 66% — Claim 1 - Nitrile-group reduction limit: not more than 46% — Claim 1 - Nitrile-group reduction limit: not more than 33% — Claim 1 - HNBR acrylonitrile content: 34 % (w/w) — Example 2 - HNBR Mooney viscosity: 63 — Example 2 - Remaining C=C double bonds: 0.8 % — Example 2 - HNBR Mn: 91.3 kg/mol — Example 2 - HNBR Mw: 297 kg/mol — Example 2 - PDI: 3.3 — Example 2 - THF-dioxane mixture ratio: 1:1 — Example 2 - Solution temperature: 50 degrees C — Example 2 - HNBR concentration: 5 % mass. — Example 2 - Co Raney particle size: 0.05-0.1 mm — Example 2 - Catalyst loading: 2.5 g — Example 2 - Substrate/catalyst ratio: 1 — Example 2 - Initial hydrogen pressure: 40 atm — Example 2 - Hydrogen pressure after heating: 100 atm — Example 2 - Reaction temperature: 220 degrees C — Example 2 - Reaction time: 20 h — Example 2 - CN-group conversion: close to 24% — Example 2

Relevant Experimental Evidence - Example 2: showed that HNBR dissolved in THF-dioxane can be further hydrogenated over Co Raney to convert nitrile groups, reaching about 24% CN conversion after 20 h. - Example 2: demonstrated that hydrogen pressure was raised from 40 atm to 100 atm after heating to 220°C. - Example 2: established that the process can tune nitrile-group saturation by varying reaction time, temperature, pressure, and mixing conditions.

Technical Relevance This patent is directly relevant because it modifies HNBR itself, converting nitrile groups to amino groups under hydrogenation conditions. It provides explicit operating windows for pressure, temperature, catalyst ratio, and time that are useful for post-hydrogenation functionalization of HNBR-derived materials.

Patent 3

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathesis-based processing of nitrile rubber in the presence of salt-activated catalyst systems, with the stated goal of improving catalyst activity in metathesis reactions including nitrile rubber degradation. The process uses a metathesis catalyst together with one or more salts, and is presented as applicable to nitrile rubber depolymerization and related transformations. - Salt: catalyst weight ratio: 0.01:1 to 10000:1 — Claim 1 - Time: 0 min — Example 7 - Time: 30 min — Example 7 - Time: 60 min — Example 7 - LiBr amount: 1 mg — Example 7 - Grubbs II catalyst amount: 10 mg — Example 7 - Chlorobenzene volume: 0.3 ml — Example 7 - CDCl₃ volume: 0.2 ml — Example 7 - Diethyl diallylmalonate addition: 0.15 ml — Example 7 - Room temperature: room temperature — Example 7 - CsBr amount: 1 mg — Example 8 - Grubbs II catalyst amount: 10 mg — Example 8 - Chlorobenzene volume: 0.3 ml — Example 8 - CDCl₃ volume: 0.2 ml — Example 8 - Diethyl diallylmalonate addition: 0.15 ml — Example 8 - Time: 15 min — Example 8 - Time: 30 min — Example 8 - Time: 60 min — Example 8 - Time: 90 min — Example 8 - Time: 150 min — Example 8

Relevant Experimental Evidence - Example 7: showed that 1 mg LiBr accelerates ring-closing metathesis of diethyl diallylmalonate relative to no-salt control. - Example 8: showed that 1 mg CsBr also accelerates ring-closing metathesis, though with a different conversion profile than LiBr. - Example 7-8: demonstrated the general salt-activated metathesis concept that underpins nitrile-rubber metathesis/degradation.

Technical Relevance This patent is relevant because it provides catalyst-system concepts for metathesis reactions of nitrile rubber, which are used in some HNBR low-Mooney routes. It supports the broader metathesis chemistry toolbox that can be coupled with hydrogenation to tailor HNBR molecular weight.

Patent 4

- Patent Number: US7666950B2
- Patent Title: Process for preparing hydrogenated nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of hydrogenated nitrile rubber by subjecting nitrile rubber to hydrogenation in a spinning disk reactor. The process is intended to reduce the hydrogen pressure burden and improve process economics relative to conventional batch hydrogenation, while using homogeneous or heterogeneous

hydrogenation catalysts. - Hydrogen pressure: 3.5 bar to 100 bar — Claim 1 - Conjugated diene content: 40 to 90 % b.w. — Detailed description - Conjugated diene content: 60 to 85 % b.w. — Detailed description - Conjugated diene content: 1,3-butadiene, isoprene, piperylene, 2,3-dimethyl butadiene — Detailed description - Unsaturated nitrile content: 10 to 60 % b.w. — Detailed description - Unsaturated nitrile content: 15 to 40 % b.w. — Detailed description - Unsaturated nitrile examples: acrylonitrile, methacrylonitrile, ethacrylonitrile — Detailed description - Additional monomer content: 0 to 40 % b.w. — Detailed description - Additional monomer content: 0.1 to 40 % b.w. — Detailed description - Additional monomer content: 1 to 30 % b.w. — Detailed description - NBR Mooney viscosity: 30 to 90 — Detailed description - NBR Mooney viscosity: 30 to 70 — Detailed description - NBR Mooney viscosity: 30 to 50 — Detailed description - NBR Mw: 200,000 to 500,000 — Detailed description - NBR Mw: 200,000 to 400,000 — Detailed description - NBR Mn: 80,000 to 150,000 — Detailed description - NBR PDI: 2.0 to 6.0 — Detailed description - NBR PDI: 2.0 to 4.0 — Detailed description - HNBR Mooney viscosity: 55 to 120 — Background - HNBR Mw: 200,000 to 500,000 — Background - HNBR PDI: greater than 3.0 — Background - Residual double bond content: up to 18% — Background - Low-Mooney HNBR Mooney viscosity: 2 to 50 — Background - Low-Mooney HNBR Mw: 30,000 to 250,000 — Background - Low-Mooney HNBR PDI: typically less than 3 — Background

Relevant Experimental Evidence - No numbered examples were detected by the parser; the patent is primarily process-description driven. - The description emphasizes that spinning disk reactors can lower hydrogen pressure requirements compared with conventional batch reactors. - The patent frames the process as a route to more economical HNBR manufacture with improved throughput and reduced equipment cost.

Technical Relevance This patent is highly relevant because it directly addresses HNBR synthesis from NBR and proposes a reactor design that can improve hydrogenation efficiency. It is important for scale-up because it links hydrogen pressure, reactor type, and polymer feed properties to HNBR production economics.

Patent 5

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a two-step metathesis-plus-hydrogenation route to low molecular weight HNBR. NBR is first subjected to olefin metathesis in the presence of ruthenium/osmium catalysts and optionally a co-olefin and solvent, then hydrogenated to give HNBR with reduced molecular weight and narrower molecular

weight distribution. - Hydrogenated nitrile rubber MW: 30,000 to 250,000 — Claim 1 - Hydrogenated nitrile rubber Mooney viscosity: 3 to 50 — Claim 1 - Hydrogenated nitrile rubber MWD/PDI: less than 2.5 — Claim 1 - Conjugated diene content: 40 to 90 % b.w. — Detailed description - Conjugated diene content: 60 to 85 % b.w. — Detailed description - Unsaturated nitrile content: 10 to 60 % b.w. — Detailed description - Unsaturated nitrile content: 15 to 40 % b.w. — Detailed description - Additional monomer content: 0 to 40 % b.w. — Detailed description - Additional monomer content: 0.1 to 40 % b.w. — Detailed description - Additional monomer content: 1 to 30 % b.w. — Detailed description - NBR Mooney viscosity: 30 to 90 — Detailed description - NBR Mooney viscosity: 30 to 70 — Detailed description - NBR Mooney viscosity: 30 to 50 — Detailed description - NBR Mw: 200,000 to 500,000 — Detailed description - NBR Mw: 200,000 to 400,000 — Detailed description - NBR Mn: 80,000 to 150,000 — Detailed description - NBR PDI: 2.0 to 6.0 — Detailed description - NBR PDI: 2.0 to 4.0 — Detailed description - Hydrogen pressure in batch hydrogenation context: 65 to 90 bar — Detailed description - Commercial HNBR Mooney viscosity: 55 to 120 — Detailed description - Commercial HNBR Mw: 200,000 to 500,000 — Detailed description - Commercial HNBR residual double bond content: up to 18% — Detailed description - Low-Mooney HNBR Mooney viscosity: 2 to 50 — Detailed description - Low-Mooney HNBR Mw: 30,000 to 250,000 — Detailed description - Low-Mooney HNBR PDI: typically less than 3 — Detailed description - Mooney viscosity of starting product Therban VP KA 8837: 55 — Detailed description - RDB of starting product Therban VP KA 8837: 18% — Detailed description - Mooney viscosity of used NBR grades: 24 to 70 — Detailed description - Mooney viscosity of used NBR grades: 28 to 40 — Detailed description - Example 1 catalyst amount: 0.05 phr — Example 1 - Example 1 catalyst amount: 0.045 phr — Example 1 - Example 1 catalyst amount: 0.060 phr — Example 1 - Example 1 metal content: 65.6 ppm — Example 1 - Example 1 metal content: 65.6 ppm — Example 1 - Example 1 metal content: 65.6 ppm — Example 1 - Example 1 co-catalyst amount: 1.0 phr triphenylphosphine — Example 1 - Example 1 reaction time: 60 min — Example 1 - Example 1 reaction time: 120 min — Example 1 - Example 1 reaction time: 180 min — Example 1 - Example 1 reaction time: 240 min — Example 1 - Example 1 RDB: 2.5% at 60 min — Example 1 - Example 1 RDB: 0.6% at 120 min — Example 1 - Example 1 RDB: 0.5% at 180 min — Example 1 - Example 1 RDB: 0.4% at 240 min — Example 1 - Example 1 HNBR 1 Mw: 113371 g/mol — Example 1 - Example 1 HNBR 2 Mw: 133758 g/mol — Example 1 - Example 1 HNBR 3 Mw: 195304 g/mol — Example 1 - Example 1 HNBR 1 Mn: 48352 g/mol — Example 1 - Example 1 HNBR 2 Mn: 54595 g/mol — Example 1 - Example 1 HNBR 3 Mn: 64130 g/mol — Example 1 - Example 1 HNBR 1 PDI: 2.34 — Example 1 - Example 1 HNBR 2 PDI: 2.45 — Example 1 - Example 1 HNBR 3 PDI: 3.05 — Example 1 - Example 1 HNBR 1 Mooney viscosity: 33.3 — Example 1 - Example 1 HNBR 2 Mooney viscosity: 41.3 — Example 1 - Example 1 HNBR 3 Mooney viscosity: 69.3 — Example 1

Relevant Experimental Evidence - Example 1: showed that the invention catalyst (HNBR 1 and HNBR 2) hydrogenates faster than the industrial norm (HNBR 3). - Example 1: demonstrated that simultaneous metathesis/hydrogenation can reduce molecular weight and Mooney viscosity of HNBR. - Example 1: showed that adding triphenylphosphine co-catalyst changes the hydrogenation profile and final polymer properties.

Technical Relevance This patent is central to low-Mooney HNBR synthesis because it combines metathesis and hydrogenation in one route to reduce molecular weight while preserving HNBR functionality. It provides a direct process window and comparative data showing how catalyst choice affects hydrogenation rate and final polymer properties.

Patent 6

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses the same low-molecular-weight HNBR concept as the related US7919563B2, using metathesis of nitrile rubber followed by hydrogenation in the presence of ruthenium/osmium catalyst systems and optionally a co-olefin. The objective is to obtain HNBR with lower molecular weight and narrower molecular weight distribution than conventional products. - Hydrogenated nitrile rubber MW: 30,000 to 250,000 — Claim 1 - Hydrogenated nitrile rubber Mooney viscosity: less than 10 — Claim 1 - Hydrogenated nitrile rubber MWD/PDI: less than 2.5 — Claim 1 - Conjugated diene content: 40 to 90 % b.w. — Detailed description - Conjugated diene content: 60 to 85 % b.w. — Detailed description - Unsaturated nitrile content: 10 to 60 % b.w. — Detailed description - Unsaturated nitrile content: 15 to 40 % b.w. — Detailed description - Additional monomer content: 0 to 40 % b.w. — Detailed description - Additional monomer content: 0.1 to 40 % b.w. — Detailed description - Additional monomer content: 1 to 30 % b.w. — Detailed description - NBR Mooney viscosity: 30 to 90 — Detailed description - NBR Mooney viscosity: 30 to 70 — Detailed description - NBR Mooney viscosity: 30 to 50 — Detailed description - NBR Mw: 200,000 to 500,000 — Detailed description - NBR Mw: 200,000 to 400,000 — Detailed description - NBR Mn: 80,000 to 150,000 — Detailed description - NBR PDI: 2.0 to 6.0 — Detailed description - NBR PDI: 2.0 to 4.0 — Detailed description - Hydrogen pressure in batch hydrogenation context: 65 to 90 bar — Detailed description - Commercial HNBR Mooney viscosity: 55 to 120 — Detailed description - Commercial HNBR Mw: 200,000 to 500,000 — Detailed description - Commercial HNBR residual double bond content: up to 18% — Detailed description - Low-Mooney HNBR Mooney viscosity: 2 to 50 — Detailed description - Low-Mooney HNBR Mw: 30,000 to 250,000 — Detailed description - Low-Mooney HNBR PDI: typically less than 3 — Detailed description - Mooney viscosity of starting product Therban VP KA 8837: 55 — Detailed description - RDB of starting product Therban VP KA 8837: 18% — Detailed description - Mooney viscosity of used NBR grades: 24 to 70 — Detailed description - Mooney viscosity of used NBR grades: 28 to 40 — Detailed description - Example 1 catalyst amount: 0.05 phr — Example 1 - Example 1 catalyst amount: 0.045 phr — Example 1 - Example 1 catalyst amount: 0.060 phr — Example 1 - Example 1 catalyst amount: 0.060 phr

1 metal content: 65.6 ppm — Example 1 - Example 1 metal content: 65.6 ppm — Example 1 - Example 1 metal content: 65.6 ppm — Example 1 - Example 1 co-catalyst amount: 1.0 phr triphenylphosphine — Example 1 - Example 1 reaction time: 60 min — Example 1 - Example 1 reaction time: 120 min — Example 1 - Example 1 reaction time: 180 min — Example 1 - Example 1 reaction time: 240 min — Example 1 - Example 1 RDB: 2.5% at 60 min — Example 1 - Example 1 RDB: 0.6% at 120 min — Example 1 - Example 1 RDB: 0.5% at 180 min — Example 1 - Example 1 RDB: 0.4% at 240 min — Example 1 - Example 1 HNBR 1 Mw: 113371 g/mol — Example 1 - Example 1 HNBR 2 Mw: 133758 g/mol — Example 1 - Example 1 HNBR 3 Mw: 195304 g/mol — Example 1 - Example 1 HNBR 1 Mn: 48352 g/mol — Example 1 - Example 1 HNBR 2 Mn: 54595 g/mol — Example 1 - Example 1 HNBR 3 Mn: 64130 g/mol — Example 1 - Example 1 HNBR 1 PDI: 2.34 — Example 1 - Example 1 HNBR 2 PDI: 2.45 — Example 1 - Example 1 HNBR 3 PDI: 3.05 — Example 1 - Example 1 HNBR 1 Mooney viscosity: 33.3 — Example 1 - Example 1 HNBR 2 Mooney viscosity: 41.3 — Example 1 - Example 1 HNBR 3 Mooney viscosity: 69.3 — Example 1

Relevant Experimental Evidence - Example 1: showed that the invention catalyst systems can produce low-Mooney HNBR with reduced molecular weight. - Example 1: compared catalyst performance and showed faster hydrogenation for the invention catalyst than the industrial benchmark. - Example 1: demonstrated that the metathesis/hydrogenation route can narrow molecular weight distribution.

Technical Relevance This patent is directly relevant because it is a core low-molecular-weight HNBR process patent using simultaneous metathesis and hydrogenation. It provides a practical route to HNBR with very low Mooney viscosity, which is valuable for improved processability and downstream compounding.

Patent 7

- Patent Number: EP2316860B1
- Patent Title: Nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses emulsion polymerization of nitrile rubber using a redox initiator system containing a peroxy compound, a reducing agent, and iron(II) chloride. It also claims nitrile rubber obtained by this method and hydrogenated nitrile rubber derived from that nitrile rubber, including HNBR made by hydrogenation or by metathesis followed by hydrogenation. - Degree of hydrogenation: 50 to 100% — Detailed description - Unsaturated nitrile content: 10 to 60 % by weight — Detailed description - Conjugated diene content: 40 to 90 % by weight — Detailed description - Conjugated

diene content: 60 to 85 % by weight — Detailed description - Conjugated diene examples: butadiene, isoprene, piperylene, 2,3-dimethylbutadiene — Detailed description - Unsaturated nitrile examples: acrylonitrile, methacrylonitrile, ethacrylonitrile — Detailed description - Additional monomer content: 0 to 40 % by weight — Detailed description - Additional monomer content: 0.1 to 40 % by weight — Detailed description - Additional monomer content: 1 to 30 % by weight — Detailed description - Mooney viscosity of NBR: 15 to 150 — Prior art discussion - Mooney viscosity of NBR: 15 to 65 — Prior art discussion - At least 0.03 mol alkylthio group per 100 mol monomer units — Prior art discussion - Temperature for polymerization in prior art: 20 to 30 degrees C — Prior art discussion - Hydrogen peroxide: oxidizing agent example — Claim 2 - Peroxodisulphates: oxidizing agent example — Claim 2 - Peroxodiphosphates: oxidizing agent example — Claim 2 - Hydroperoxides: oxidizing agent example — Claim 2 - Peracids: oxidizing agent example — Claim 2 - Peracid esters: oxidizing agent example — Claim 2 - Peracid anhydrides: oxidizing agent example — Claim 2 - Peroxides having two organic radicals: oxidizing agent example — Claim 2 - Sodium, potassium or ammonium salts of peroxodisulphuric acid: oxidizing agent example — Claim 3 - Sodium, potassium or ammonium salts of peroxodiphosphoric acid: oxidizing agent example — Claim 3 - t-Butyl hydroperoxide: oxidizing agent example — Claim 3 - Cumene hydroperoxide: oxidizing agent example — Claim 3 - Pinane hydroperoxide: oxidizing agent example — Claim 3 - p-Menthane hydroperoxide: oxidizing agent example — Claim 3 - Dibenzoyl peroxide: oxidizing agent example — Claim 3 - 2,4-dichlorobenzoyl peroxide: oxidizing agent example — Claim 3 - Di-t-butyl peroxide: oxidizing agent example — Claim 3 - Dicumyl peroxide: oxidizing agent example — Claim 3 - t-Butyl perbenzoate: oxidizing agent example — Claim 3 - t-Butyl peracetate: oxidizing agent example — Claim 3

Relevant Experimental Evidence - No numbered examples were detected by the parser; the patent is primarily directed to polymerization chemistry and product claims. - The patent emphasizes that redox initiation with Fe(II) chloride is used to make nitrile rubber precursors for subsequent hydrogenation. - It explicitly claims hydrogenated nitrile rubber obtainable by hydrogenation or by metathesis followed by hydrogenation.

Technical Relevance This patent is relevant because it defines the NBR precursor synthesis route that feeds into HNBR manufacture. The redox-emulsion polymerization conditions and monomer ranges are important for controlling precursor composition, which strongly influences hydrogenation behavior and final HNBR properties.

Patent 8

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed

- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses hydrogenation of NBR to HNBR in the presence of ruthenium, palladium, or rhodium compounds, with a strong emphasis on catalyst residue control to improve aging properties. The process is carried out in solution and is designed to avoid separate catalyst-removal steps while still producing HNBR with low discoloration and improved stability. - Conjugated diene content: 40 to 90 % by weight — Claim 1 - Unsaturated nitrile content: 10 to 60 % by weight — Claim 1 - Ruthenium amount: 10 to 200 ppm — Claim 1 - Ruthenium amount: 10 to 150 ppm — Claim 1 - Ruthenium amount: 10 to 120 ppm — Claim 1 - Ruthenium amount: 10 to 79 ppm — Claim 1 - Ruthenium amount: 42 to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm — Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200 degrees C — Claim 1 - Hydrogenation pressure: 700000 to 15000000 pascals — Claim 1 - Hydrogenation time: 1 to 24 h — Claim 1 - Ageing criterion: gel content increase less than 10% after 4 days at 140 degrees C — Detailed description - Ageing criterion: Mooney viscosity increase less than 40% after 4 days at 140 degrees C — Detailed description - HNBR rhodium content: 58 to 60 ppm — Prior art discussion - Hydrogenation catalyst loading: 0.45 parts magnesium-silicate-supported Pd catalyst — Prior art discussion - Rh content after purification: 5.8 to 9.2 ppm — Prior art discussion - Removal rate: 99% — Prior art discussion

Relevant Experimental Evidence - No numbered examples were detected by the parser; the patent is largely focused on catalyst residue management and product-aging performance. - The patent argues that noble-metal residues darken HNBR and worsen aging and gel content. - It surveys multiple catalyst-removal technologies and positions the invention as a route to better aging without a separate removal step.

Technical Relevance This patent is highly relevant because it directly addresses HNBR hydrogenation and the practical issue of residual catalyst contamination. It provides explicit catalyst loading and process windows that are important for producing aged-stable HNBR with acceptable color and gel behavior.

Patent 9

- Patent Number: EP2473280B1
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of nitrile rubber in the presence of hydrogen and a ruthenium/osmium catalyst system, optionally with a co-olefin. The process is intended to produce HNBR with lower molecular weight, lower Mooney viscosity, and narrower molecular weight distribution in a single step. - Hydrogenated nitrile rubber MW: 20,000 to 250,000 — Detailed description - Hydrogenated nitrile rubber Mooney viscosity: 1 to 50 — Detailed description - Hydrogenated nitrile rubber MWD/PDI: less than 2.8 — Detailed description - NBR Mooney viscosity: 24 to 70 — Detailed description - NBR Mooney viscosity: 28 to 40 — Detailed description - Example 1 catalyst amount: 0.057 phr Grubbs III — Example 1 - Example 1 catalyst amount: 0.080 phr Grubbs III — Example 1 - Example 1 catalyst amount: 0.06 phr Wilkinson's catalyst — Example 1 - Example 1 catalyst amount: 0.055 phr Grubbs II — Example 1 - Example 1 metal content: 65.6 ppm — Example 1 - Example 1 metal content: 91.3 ppm — Example 1 - Example 1 metal content: 65.6 ppm — Example 1 - Example 1 reaction time: 0 min — Example 1 - Example 1 reaction time: 60 min — Example 1 - Example 1 reaction time: 120 min — Example 1 - Example 1 reaction time: 180 min — Example 1 - Example 1 reaction time: 240 min — Example 1 - Example 1 RDB: 100% at 0 min — Example 1 - Example 1 RDB: 2.0% at 60 min — Example 1 - Example 1 RDB: 0.4% at 120 min — Example 1 - Example 1 RDB: 0.3% at 180 min — Example 1 - Example 1 RDB: 0.7% at 60 min for Wilkinson's catalyst — Example 1 - Example 1 RDB: 0.6% at 120 min for Wilkinson's catalyst — Example 1 - Example 1 RDB: 0.3% at 180 min for Wilkinson's catalyst — Example 1 - Example 1 HNBR 1 Mw: 191541 g/mol — Example 1 - Example 1 HNBR 2 Mw: 135772 g/mol — Example 1 - Example 1 HNBR 3 Mw: 195304 g/mol — Example 1 - Example 1 HNBR 4 Mw: 144212 g/mol — Example 1 - Example 1 HNBR 1 Mn: 68643 g/mol — Example 1 - Example 1 HNBR 2 Mn: 58259 g/mol — Example 1 - Example 1 HNBR 3 Mn: 64130 g/mol — Example 1 - Example 1 HNBR 4 Mn: 59851 g/mol — Example 1 - Example 1 HNBR 1 PDI: 2.79 — Example 1 - Example 1 HNBR 2 PDI: 2.33 — Example 1 - Example 1 HNBR 3 PDI: 3.05 — Example 1 - Example 1 HNBR 4 PDI: 2.45 — Example 1 - Example 1 HNBR 1 Mooney viscosity: 38.2 — Example 1 - Example 1 HNBR 2 Mooney viscosity: 20.2 — Example 1 - Example 1 HNBR 3 Mooney viscosity: 69.3 — Example 1 - Example 1 HNBR 4 Mooney viscosity: 32.6 — Example 1

Relevant Experimental Evidence - Example 1: compared Grubbs III, Grubbs II, and Wilkinson's catalyst for simultaneous hydrogenation/metathesis of NBR. - Example 1: showed that Grubbs III hydrogenates faster than Grubbs II or Wilkinson's catalyst at equal metal content. - Example 1: demonstrated that the process can reduce Mooney viscosity and molecular weight while lowering residual double bonds.

Technical Relevance This patent is one of the most directly relevant disclosures for HNBR synthesis because it combines hydrogenation and metathesis to reach low-Mooney products in a single step. The comparative catalyst data are especially useful for selecting catalyst systems and process conditions for controlled HNBR manufacture.

Patent 10

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber

- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a process for producing low molecular weight HNBR by contacting nitrile rubber with a ruthenium or osmium catalyst and carrying out simultaneous hydrogenation and metathesis in the presence of hydrogen and optionally a co-olefin. The process is positioned as an alternative to prior Grubbs-type systems, with the goal of obtaining lower molecular weight and narrower molecular weight distribution HNBR. - Hydrogenated nitrile rubber MW: 20,000 to 250,000 — Detailed description - Hydrogenated nitrile rubber Mooney viscosity: 1 to 50 — Detailed description - Hydrogenated nitrile rubber MWD/PDI: less than 3.0 — Detailed description - NBR Mooney viscosity: 24 to 70 — Detailed description - NBR Mooney viscosity: 28 to 40 — Detailed description - Example 1 catalyst amount: 0.041 phr Hoveyda — Example 1 - Example 1 catalyst amount: 0.045 phr Arlt — Example 1 - Example 1 catalyst amount: 0.060 phr Wilkinson's catalyst — Example 1 - Example 1 metal content: 65.6 ppm — Example 1 - Example 1 metal content: 65.6 ppm — Example 1 - Example 1 metal content: 65.6 ppm — Example 1 - Example 1 co-catalyst amount: NIL — Example 1 - Example 1 co-catalyst amount: NIL — Example 1 - Example 1 co-catalyst amount: 1.0 phr triphenylphosphine — Example 1 - Example 1 reaction time: 0 min — Example 1 - Example 1 reaction time: 60 min — Example 1 - Example 1 reaction time: 120 min — Example 1 - Example 1 reaction time: 180 min — Example 1 - Example 1 reaction time: 240 min — Example 1 - Example 1 RDB: 100% at 0 min — Example 1 - Example 1 RDB: 2.5% at 60 min — Example 1 - Example 1 RDB: 0.6% at 120 min — Example 1 - Example 1 RDB: 0.5% at 180 min — Example 1 - Example 1 RDB: 0.4% at 240 min — Example 1 - Example 1 HNBR 1 Mw: 113371 g/mol — Example 1 - Example 1 HNBR 2 Mw: 133758 g/mol — Example 1 - Example 1 HNBR 3 Mw: 195304 g/mol — Example 1 - Example 1 HNBR 1 Mn: 48352 g/mol — Example 1 - Example 1 HNBR 2 Mn: 54595 g/mol — Example 1 - Example 1 HNBR 3 Mn: 64130 g/mol — Example 1 - Example 1 HNBR 1 PDI: 2.34 — Example 1 - Example 1 HNBR 2 PDI: 2.45 — Example 1 - Example 1 HNBR 3 PDI: 3.05 — Example 1 - Example 1 HNBR 1 Mooney viscosity: 33.3 — Example 1 - Example 1 HNBR 2 Mooney viscosity: 41.3 — Example 1 - Example 1 HNBR 3 Mooney viscosity: 69.3 — Example 1

Relevant Experimental Evidence - Example 1: compared Hoveyda, Arlt, and Wilkinson's catalysts for simultaneous hydrogenation/metathesis of NBR. - Example 1: showed that the invention catalysts outperform the industrial norm in hydrogenation efficiency. - Example 1: demonstrated low-Mooney HNBR formation with reduced molecular weight and narrower PDI.

Technical Relevance This patent is directly relevant because it targets low molecular weight HNBR through a single-step hydrogenation/metathesis process. It provides practical catalyst comparisons and confirms that catalyst selection strongly affects both hydrogenation rate and final polymer properties.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2473280B1 and US9150669B2 both use simultaneous hydrogenation/metathesis of NBR with ruthenium/osmium catalyst systems, but EP2473280B1 reports a slightly broader low-Mooney target window (Mooney 1-50, PDI <2.8) and US9150669B2 reports Mooney 1-50 with PDI <3.0.
- US7919563B2 and US7772328B2 both pursue low-Mooney HNBR via metathesis followed by hydrogenation; US7919563B2 claims PDI <2.5 while US7772328B2 claims PDI <2.5 and a stricter Mooney viscosity of less than 10.
- US7666950B2 is process-engineering focused, using a spinning disk reactor and hydrogen pressures of 3.5-100 bar, whereas the metathesis-based patents rely on ruthenium/osmium catalyst chemistry rather than reactor intensification.
- EP3116916B1 is distinct from the other patents because it does not primarily make HNBR from NBR; instead it reduces nitrile groups on already hydrogenated rubber using Raney cobalt at 170-220°C and 0.5-25 MPa H₂.
- EP3827029A1 emphasizes catalyst residue control and aging performance, with Ru/Pd/Rh loadings in the tens to hundreds of ppm, while the other hydrogenation patents focus more on molecular-weight control or reactor/catalyst selection.
- EP2316860B1 addresses precursor NBR synthesis by emulsion polymerization with Fe(II) chloride redox initiation, providing the feedstock platform for downstream HNBR hydrogenation rather than the hydrogenation step itself.
- Across the hydrogenation-focused patents, solution-phase processing is common, but the process windows differ: EP3827029A1 uses 60-200°C and 0.7-15 MPa, EP3116916B1 uses 170-220°C and 0.5-25 MPa, and US7666950B2 broadens hydrogen pressure to 3.5-100 bar in a spinning disk reactor.
- Catalyst systems vary substantially: Grubbs/Hoveyda/Wilkinson-type metathesis catalysts dominate the low-Mooney routes, while noble-metal hydrogenation catalysts and Raney cobalt dominate direct hydrogenation or post-functionalization routes.

4. CONCLUSION

The evidence shows that HNBR synthesis is most effectively advanced by two complementary strategies: direct hydrogenation of NBR under carefully controlled catalyst and reactor conditions, and metathesis-assisted routes that lower molecular weight before or during hydrogenation. For low-Mooney HNBR, the strongest evidence supports simultaneous hydrogenation/metathesis using ruthenium/osmium catalyst systems, with catalyst loadings in the 0.04-0.08 phr range and reaction times on the order of 1-4 h producing substantial reductions in residual double bonds and molecular weight. For conventional hydrogenation, solution-phase operation with noble-metal catalysts remains important, but catalyst residue control is critical for aging and color stability.

Reactor intensification via spinning disk reactors and careful hydrogen-pressure management can reduce equipment burden and improve process economics. If the goal is post-functionalized HNBR, Raney cobalt hydrogenation of nitrile groups is a viable route, but it requires high temperature and pressure and only partial nitrile reduction. Overall, the most evidence-supported synthesis parameters are: controlled NBR precursor composition, low-to-moderate Mooney feedstock, ruthenium-based metathesis/hydrogenation catalysts for low-Mooney products, and rigorous management of catalyst residues when noble-metal hydrogenation is used.

5. REFERENCES

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- EP3116916B1 | Process for the reduction of nitrile groups of hydrogenated nitrile rubber | Not disclosed | EP | 2018 | <https://patents.google.com/patent/EP3116916B1/en>
- US8957161B2 | Process for reacting chemical compounds in the presence of catalyst systems | Not disclosed | US | 2015 | <https://patents.google.com/patent/US8957161B2/en>
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- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>